

ITHACA 13 E, NY, USA

WMO: 725270

Lat: 42.4401N

Lon: 76.2462W

Elev: 374

StdP: 96.91

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 64758

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.1	(c) -17.1	(d) -23.9	(e) 0.5	(f) -20.0	(g) -21.4	(h) 0.6	(i) -17.1	(j) 8.3	(k) -5.3	(l) 7.6	(m) -5.9	(n) 1.6	(o) N/A	(p) 0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.8	(c) 29.8	(d) 21.4	(e) 28.4	(f) 20.7	(g) 26.9	(h) 20.2	(i) 23.0	(j) 27.8	(k) 22.1	(l) 26.4	(m) 21.3	(n) 25.3	(o) 2.5	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	16.8	25.6	20.7	16.0	24.6	20.0	15.3	23.9	70.0	28.1	66.6	26.6	63.7	25.1	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.4	(b) 6.6	(c) 5.8	(e) -24.7	(f) 32.3	(g) 3.5	(h) 1.6	(i) -27.2	(j) 33.4	(k) -29.2	(l) 34.4	(m) -31.2	(n) 35.3	(o) -33.8	(p) 36.5		
(a) 7.4	(b) 6.6	(c) 5.8	(e) -25.5	(f) 24.4	(g) 3.1	(h) 0.8	(i) -27.8	(j) 24.9	(k) -29.6	(l) 25.4	(m) -31.4	(n) 25.8	(o) -33.6	(p) 26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-5.3	-5.1	-0.4	6.7	13.0	17.5	20.0	18.8	15.4	9.6	3.2	-1.9		
	DBStd	10.25	6.76	6.06	6.14	5.05	4.81	3.55	2.99	3.08	4.07	4.68	5.13	5.32		
	HDD10.0	2020	476	422	327	122	24	1	0	0	3	67	209	369		
	HDD18.3	4063	734	655	581	348	177	57	18	33	107	274	453	627		
	CDD10.0	1178	0	0	4	24	117	227	309	272	166	53	6	1		
	CDD18.3	181	0	0	0	1	11	33	68	46	19	2	0	0		
	CDH23.3	1589	0	0	0	22	137	306	602	354	157	11	0	0		
	CDH26.7	347	0	0	0	4	24	71	150	67	30	1	0	0		
(14) Wind	WSAvg	2.6	3.1	3.3	3.3	3.1	2.4	2.1	1.8	1.8	1.9	2.4	2.8	3.2		
(15) Precipitation	PrecAvg	1028	68	56	82	84	84	109	101	102	98	98	72	75		
	PrecMax	1247	122	116	155	202	170	231	224	199	174	199	110	139		
	PrecMin	791	30	28	31	21	31	35	38	47	55	30	26	35		
	PrecStd	137	25	23	31	39	38	47	47	38	35	49	24	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.0	14.4	21.0	26.3	29.2	30.8	32.0	30.6	29.8	25.0	18.8	15.4		
		MCWB	12.5	10.1	13.8	17.8	20.2	22.0	22.6	22.1	22.0	18.6	13.1	12.9		
	2%	DB	9.9	9.5	15.1	22.3	27.0	28.8	29.9	28.6	27.2	21.8	15.8	10.4		
		MCWB	7.9	5.9	10.3	14.7	19.2	20.1	21.7	21.3	20.8	16.9	11.6	8.9		
	5%	DB	6.6	5.9	11.9	18.9	24.7	26.8	28.4	26.8	25.2	19.8	13.7	7.8		
		MCWB	4.7	3.1	8.4	12.1	18.0	19.9	20.9	20.4	19.9	15.9	10.0	6.2		
	10%	DB	3.8	3.1	8.8	16.2	22.3	24.9	26.8	25.3	22.8	17.5	11.4	5.5		
		MCWB	2.4	0.8	5.8	11.2	16.2	18.5	20.3	19.7	18.7	14.4	8.3	4.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	13.1	15.1	17.7	21.9	22.9	24.3	23.8	23.5	20.3	14.3	13.3		
		MCDB	12.3	16.6	20.8	22.6	26.7	28.8	30.2	27.7	28.2	23.2	16.1	15.6		
	2%	WB	7.8	7.5	12.2	15.1	20.6	21.7	23.2	22.5	21.9	18.4	12.5	10.0		
		MCDB	8.9	10.9	16.7	19.2	25.6	26.5	28.5	26.3	25.6	20.5	15.6	11.2		
	5%	WB	4.9	4.1	9.4	13.0	19.1	20.7	22.3	21.4	20.8	17.0	10.5	7.3		
		MCDB	6.1	6.6	12.1	16.7	23.3	24.8	27.0	25.0	24.9	19.4	13.0	8.4		
	10%	WB	2.6	1.5	6.6	11.3	17.7	19.6	21.4	20.5	19.4	15.1	8.5	5.0		
		MCDB	4.0	3.4	9.5	14.9	21.4	23.2	25.5	23.8	22.2	17.4	11.3	6.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	9.0	9.8	11.5	12.4	11.4	11.8	11.3	11.5	9.7	8.9	7.2		
		MCDBR	10.4	12.1	14.0	17.5	15.5	14.1	13.7	12.7	13.4	12.8	12.5	9.4		
	5% WB	MCWBR	9.6	9.3	9.4	9.9	8.0	7.6	6.5	6.6	7.0	8.0	8.6	7.9		
		MCDWR	10.1	12.5	12.8	12.9	12.3	11.7	11.4	10.7	11.4	10.4	10.6	8.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.291	0.309	0.330	0.357	0.384	0.434	0.441	0.423	0.362	0.347	0.305	0.285		
		taud	2.375	2.328	2.389	2.373	2.336	2.213	2.207	2.244	2.401	2.447	2.517	2.472		
	Ebn at Noon	Ebn at Noon	850	891	915	911	890	842	832	835	869	839	836	833		
		Edh at Noon	80	99	106	116	124	140	140	131	104	88	70	67		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.42	2.29	3.37	4.37	5.24	5.55	5.71	5.08	4.07	2.55	1.65	1.12		
	RadStd	RadStd	0.12	0.32	0.30	0.51	0.52	0.52	0.44	0.29	0.41	0.28	0.21	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+93	N/A

Nomenclature: See separate page